Task 4: Agreement Curve for K-values

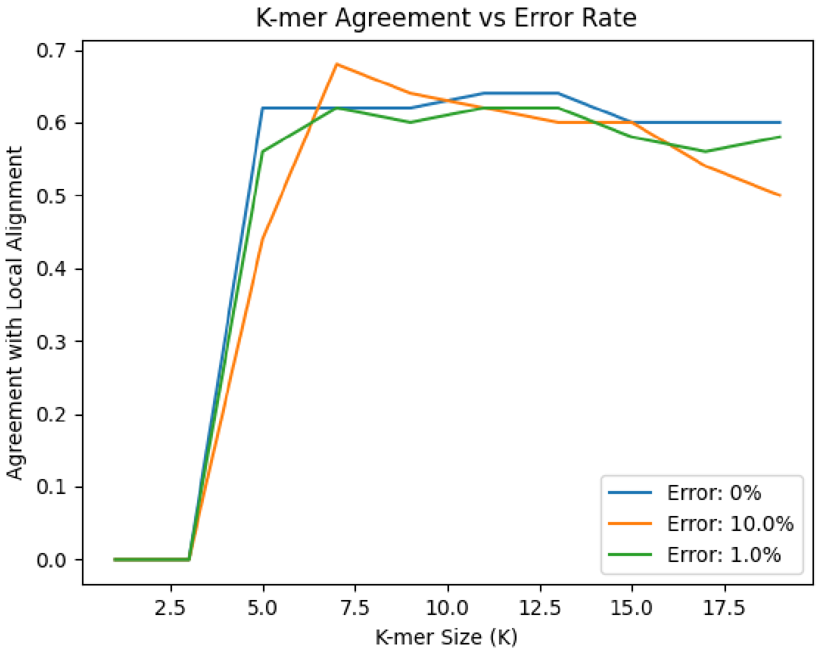

Figure 1: Agreement with local alignment with different length kmers. The orange line is with 10% mutation on the query set (simulating Oxford Nanopore). The green line is with 1% mutation on the query set (simulating Illumina).

Interestingly, the agreement with local alignment has a higher maximum for the mutated dataset. This is not too surprising, as the baseline that we are comparing this to is local aligning upon the mutated dataset as well. This might suggest that kmer matching is more robust against sequencing errors than local alignment.

Task 8: Computation time benchmark

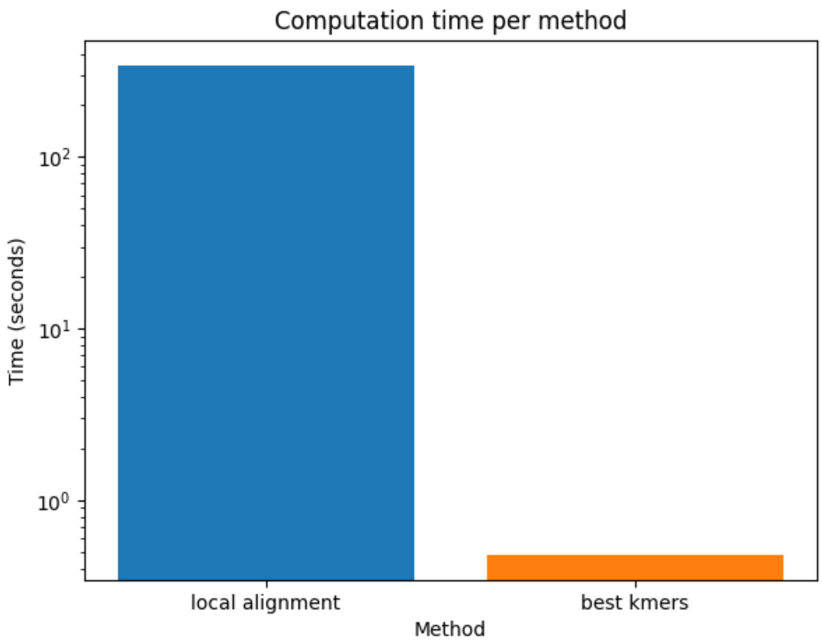

Figure 2: The amount of time each method took to precompute with time plotted on a log scale. Note that the kmer method was under a second and the local alignment took about 5 minutes. All code was run with pypy3.

The timing results show that running local alignment to match sequences can be very costly. The kmer version, however, is very quick in comparison. This means that sometimes it may be useful to sacrifice the accuracy of the local alignment for the speed of comparing kmers.

Task 9: Blast Results

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenBank Graphics Distance tree of results MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Bacillus cereus strain AFS040984 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1535	OP986665.1
Bacillus sp. (in: Bacteria) strain EDA4 16S ribosomal RNA gene, partial sequence	Bacillus sp. (in: firmicutes)	2603	2603	100%	0.0	100.00%	1485	MZ348870.1
Bacillus thuringiensis strain NBAIR-BT25 16S ribosomal RNA gene, partial sequence	Bacillus thuringiensis	2603	2603	100%	0.0	100.00%	1516	MN327970.1
Bacillus cereus strain HBUAS74157 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1493	PQ326546.1
Bacillus cereus strain HTI605-1 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1455	MG835999.1
Bacillus cereus strain FHP 1161 chromosome	Bacillus cereus	2603	33769	100%	0.0	100.00%	5096982	CP185782.1
Bacillus cereus strain Z009 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1507	MG266470.1
Bacillus thuringiensis strain JJ1216 chromosome, complete genome	Bacillus thuringiensis	2603	36226	100%	0.0	100.00%	5363548	CP160237.1
Bacillus thuringiensis strain BM-BT15426, complete genome	Bacillus thuringiensis	2603	36259	100%	0.0	100.00%	5246329	CP020723.1
Bacillus thuringiensis strain IMBL-B9 chromosome, complete genome	Bacillus thuringiensis	2603	33760	100%	0.0	100.00%	5620474	CP072691.1
Bacillus cereus strain FDAARGOS_799 chromosome, complete genome	Bacillus cereus	2603	36298	100%	0.0	100.00%	5381556	CP053951.1
Bacillus cereus strain A8 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1492	OQ940476.1
Bacillus cereus strain IBA1 chromosome, complete genome	Bacillus cereus	2603	33767	100%	0.0	100.00%	5702443	CP106987.1
Bacillus cereus strain BTCB20 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1487	MK118714.1
Bacillus cereus strain HC07-06-qm 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1486	OM533609.1
Bacillus cereus strain WSK13 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1492	KY085975.1
Bacillus cereus strain Z005 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1507	MG255969.1
Bacillus cereus strain TFBc04 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1515	OQ584437.1
Bacillus sp. (in: Bacteria) strain HVul.ww2 16S ribosomal RNA gene, partial sequence	Bacillus sp. (in: firmicutes)	2603	2603	100%	0.0	100.00%	1496	MW092227.1
Bacillus cereus strain KSW-91 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1488	OR514188.1
Bacillus thuringiensis strain IPPBIOTSUC-1012 chromosome, complete genome	Bacillus thuringiensis	2603	36337	100%	0.0	100.00%	5476282	CP170495.1
Bacillus cereus strain FORC_013 chromosome, complete genome	Bacillus cereus	2603	36403	100%	0.0	100.00%	5418913	CP011145.1
Bacillus cereus strain PHAG6 16S ribosomal RNA gene, partial sequence	Bacillus cereus	2603	2603	100%	0.0	100.00%	1474	PX796478.1

Figure 3: The blast results given from blastn from the https://blast.ncbi.nlm.nih.gov/Blast.cgi website. Results are ordered by how well they are matched. The sequence run was the C12 sequence.

From the blastn results we can see that the C12 sequence matches Bacillus Cereus the best. This bacteria grows in various natural places including soil and food. This is a bit surprising given that it was gathered from the door to the library. Perhaps many people eat food before going to the library and this bacteria is from their hands. When we grew this on our plate we had a circular shape, which is consistent with what this tends to grow as.

Task 10: Alignment results

I used local alignment with my database on the same C12 sequence and got a match for Streptococcus. This is different from the earlier result, perhaps because the database I have does not contain this bacteria in it. However, this is quite strange as these bacteria are quite different.